

Sarus Field Test Checklist

Nothing here is optional. Work through the four phases in order before Sarus Operation Planner controls a real aircraft, and if an item fails, stop there and find out why before going on. The ticks are remembered by this browser only, so use one device per test.

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PROPS OFF

Every propeller stays off through phases A, B and C. They go on for phase D only, outdoors, with a second person watching and the pilot holding the RC transmitter.

A Before you start

At the desk, before connecting anything.

☐

Record the versions under test

Sarus version (Help screen), flight controller board, ArduPilot firmware version, telemetry radio model.

☐

Save a parameter backup

CONFIG → Full Parameter List → **Save**. Keep the file with the test record.

☐

Check the RC transmitter failsafe

Transmitter off → receiver outputs failsafe throttle. Know which flight mode the aircraft enters.

☐

Choose the light and dark theme for the field

Open Sarus outdoors in sunlight and pick the theme you can read best.

B Connect

Aircraft on the bench with the battery in. Props still off.

☐

Connect over USB

Pick the COM port, 115200, Connect. **Expect:** parameters download, HUD moves, title bar shows the firmware version.

☐ **Connect over the telemetry radio**

Usually 57600. **Expect:** link quality above 90% at bench distance, no "no data" warnings.

☐ **Unplug and replug the radio while connected**

Expect: telemetry resumes within a few seconds of reconnecting, no error dialogs.

c **Bench checks**

Props off. For these checks you move the aircraft by hand and watch the screen.

☐ **Attitude matches the airframe**

Roll right, pitch nose up, yaw right by hand. **Expect:** the HUD moves the same way.

☐ **Heading and GPS**

Expect: heading within about 10° of a phone compass; outdoors, 3D fix with HDOP below 1.5.

☐ **Battery reading**

Compare with a multimeter. **Expect:** voltage within 0.1 V.

☐ **RC inputs and flight mode switch**

SETUP → Radio Calibration. Move every stick and switch. **Expect:** the mode shown in Sarus matches the switch position.

☐ **Motor and servo test**

SETUP → Optional Hardware → Motor Test at 5% throttle, props off. **Expect:** each VTOL motor spins in the order and direction of your frame diagram; control surfaces move the right way.

☐ **Parameter write and read-back**

Change TECS_CLMB_MAX by 0.5 in the Full Parameter List, Write, then set it back. **Expect:** no "values changed by aircraft" message for a normal value.

☐ **Out-of-range warning**

Type a value above the recommended maximum. **Expect:** amber cell with the range in its tooltip and no blocking question. Do not write it; press Refresh.

☐ **Mission, fence and rally round trip**

PLAN: create a short mission with VTOL takeoff and VTOL land, Write, then Read. **Expect:** identical points, altitudes and commands. Repeat for a geofence and a rally point.

☐ **Arming checks are shown**

Try to arm with a known problem (for example no GPS fix). **Expect:** the PreArm message appears on the HUD and in Messages.

☐ **Log download**

DATA → DataFlash Logs → Download. **Expect:** the latest log downloads and opens in Review a Log.

☐ **Telemetry link loss on the bench**

Power off the ground radio for 15 s, then on. **Expect:** Sarus shows the loss, telemetry returns, and the aircraft takes the GCS failsafe action you configured.

D First flight, QuadPlane

An open area, props on, an observer beside you and the transmitter ready to take over. Go higher or further only when the step before has worked.

☐ **QHOVER hover at 3–5 m**

Expect: stable hover; Sarus altitude within about 1 m of a visual estimate.

☐ **QLOITER hold**

Expect: holds position within a few metres for 30 s; mode shown correctly in Sarus.

☐ **Transition to fixed wing**

At a safe height, switch to FBWA. **Expect:** "Transition" messages in Sarus, airspeed rising past ARSPD_FBW_MIN, clean forward flight.

☐ **TECS behaviour**

FBWB or CRUISE, then climb and descend. **Expect:** climb and sink rates close to TECS_CLMB_MAX and TECS_SINK_MAX.

☐ **Small AUTO mission**

VTOL takeoff, two waypoints, VTOL land. **Expect:** Sarus shows the current waypoint and the aircraft lands within a few metres of the planned point.

☐ **Return home**

Switch to RTL (or QRTL) from a distance. **Expect:** the aircraft returns and lands as configured.

☐ **Link loss at altitude**

With the pilot ready on the transmitter, switch off the ground radio for 10 s. **Expect:** the configured GCS failsafe action; Sarus recovers telemetry when the radio is back on.

Keep with the test record

File the telemetry logs (.tlog, found in Documents\Sarus Operation Planner\logs) and the flight controller's .bin log for every flight, together with the parameter backup from step A2. Add a note for anything unexpected, a message or a behaviour, and write down when it happened. A time stamp makes the log far easier to read later.

Sarus Aerospace Pvt Ltd · checklist for Sarus Operation Planner 1.0.0. Heights, distances and failsafe actions here are examples; set them for your aircraft and for the Drone Rules that apply where you fly.